

Roles & Titles — module PRD

Build spec for the role taxonomy module of the Executive Engine: **Role group** → **Role** → **Title variant**, with favorite titles promoted through discovery, scoring, queueing, and notification. Reference implementation: `proto-compass/roles.jsx` + `proto-compass/taxonomy.js` in `Executive Engine Prototype (Compass).html` (route `#/roles`). Design system: Compass (teal/slate, Inter, 13–14px base).

How to use this document. Every screenshot below is a real capture of the reference implementation, labelled with the exact route, the DOM hook that triggers it, and the expected result. Every interactive element carries a stable `data-*` attribute — build against those names and the click-through is 1:1. Screenshots were taken at 1560×910 (three-pane), 1000×580 (two-pane) and 620×940 (stacked).

1. Problem & goal

The engine watches mailbox folders for job alerts. Today it matches on a flat list of eight role "personas", so a Chief Digital & Technology Officer post and a fintech CTO post score identically. The user's search is **title-shaped**, **not role-shaped**: within one role (CTO) some variants are exactly the target (Platform CTO, Enterprise CTO) and others are noise (Field CTO, Fractional CTO).

Goal: model the taxonomy three levels deep, let the user mark *favorite* titles, and make that favorite signal do real work — score, queue order, card treatment, alert timing, packet autostart. Secondary goal: replace the hardcoded folder map with a per-role folder binding.

Success criteria

- A user can find any of ~484 seeded titles in under 5 seconds (tree + search).
- Marking a title favorite changes its opportunities' rank without any other action.
- Turning a title off means it never appears in the swipe queue again.
- Inclusion rules (COO, Director) are enforced by the matcher, not by the user's memory.

2. Domain concepts

CONCEPT	DEFINITION	CARDINALITY
Role group	Seniority band the user searches in. Seeded: <code>C Suite</code> , <code>VP & Head of</code> , <code>Director</code> . Carries an optional group-level inclusion rule and a fallback watched folder.	3 seeded, user-extensible
Role	A function within a group — <code>CTO</code> , <code>CIO</code> ... in C Suite; <code>Software</code> , <code>Engineering</code> , <code>Product</code> ... in VP and Director. Owns the watched folder, the inclusion rule, and the role baseline (resume/narrative/comp).	27 seeded (7 + 10 + 10)

Title variant	A literal job title string the matcher looks for, including long and abbreviated forms ("Chief Technology Officer" and "CTO" are two variants). Each carries a tier.	484 seeded
Tier	★ favorite desired — promoted · watch default — ingested normally · off excluded from ingestion entirely.	enum
Inclusion rule	Machine-checkable predicate + human text, attached to a group or role. Two seeded: COO (requires software/digital-transformation/product-delivery/tech-ops/business-transformation centrality) and Director (requires Senior/Executive/Managing seniority, or an exceptional-scope exception).	0..1 per role/group
Folder binding	Role → mailbox folder to watch. Falls back to the group folder when unset.	1 per role
Role baseline	Existing concept — resume/narrative/comp reference the packet builder tailors from. Now hangs off Role.	0..1 per role

3. Data schema

Postgres. All user-scoped tables carry `user_id`; taxonomy is per-user (seeded from a global template on signup) so the user can rename, add, and reorder freely.

```
-- — Taxonomy —————
create type tier_t as enum ('fav','watch','off');

create table role_group (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid not null references app_user(id) on delete cascade,
  name        text not null,           -- 'C Suite'
  slug        text not null,           -- 'csuite'
  seniority   text,                    -- 'C-level' | 'VP / Head of' | 'Senior / Executive /
Managing Director'
  folder_id   uuid references mail_folder(id), -- fallback watched folder
  rule_text   text,                    -- human-readable inclusion rule
  rule        jsonb,                   -- machine predicate, see §5
  sort_order  int not null default 0,
  unique (user_id, slug)
);

create table role (
  id          uuid primary key default gen_random_uuid(),
  group_id    uuid not null references role_group(id) on delete cascade,
  name        text not null,           -- 'CTO' | 'Data, Analytics & AI'
  slug        text not null,           -- 'cto' | 'vp-data'
  folder_id   uuid references mail_folder(id),
  rule_text   text,
  rule        jsonb,
  baseline_id uuid references role_baseline(id),
  sort_order  int not null default 0,
```

```

    unique (group_id, slug)
);

create table title_variant (
  id          uuid primary key default gen_random_uuid(),
  role_id     uuid not null references role(id) on delete cascade,
  label       text not null,           -- 'Platform CTO' (display, verbatim)
  normalized  text not null,           -- 'platform chief technology officer' (see §5)
  tier         tier_t not null default 'watch', -- PUBLISHED tier
  is_seed     boolean not null default true,
  sort_order  int not null default 0,
  unique (role_id, label)
);

create index on title_variant (normalized);
create index on title_variant (role_id, tier);

-- Draft layer: the working set behind "Save favorites" / "Revert N"
create table title_tier_draft (
  user_id     uuid not null references app_user(id) on delete cascade,
  title_id    uuid not null references title_variant(id) on delete cascade,
  tier         tier_t not null,
  updated_at  timestamptz not null default now(),
  primary key (user_id, title_id)
);

-- — Mailbox —————
create table mail_connection (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid not null references app_user(id) on delete cascade,
  provider     text not null,           -- 'google' | 'microsoft' | 'yahoo'
  email       text not null,
  scope       text not null default 'readonly',
  status      text not null default 'active',
  last_scan_at timestamptz
);

create table mail_folder (
  id          uuid primary key default gen_random_uuid(),
  connection_id uuid not null references mail_connection(id) on delete cascade,
  provider_path text not null,         -- 'Jobs/CTO'
  display_name text not null,
  unique (connection_id, provider_path)
);

-- — Matching output —————
create table title_match (
  id          uuid primary key default gen_random_uuid(),
  alert_id    uuid not null references mail_alert(id) on delete cascade,
  opportunity_id uuid references opportunity(id) on delete cascade,
  title_id    uuid references title_variant(id),
  role_id     uuid references role(id),
  method      text not null,           -- 'exact' | 'normalized' | 'fuzzy' | 'none'
  confidence  numeric(4,3) not null,
  tier_at_match tier_t,
  rule_passed boolean not null default true,

```

```

rule_reason    text,
created_at     timestampz not null default now()
);

alter table opportunity
  add column matched_title_id uuid references title_variant(id),
  add column role_id          uuid references role(id),
  add column base_score       int,      -- 0..100 before promotion
  add column tier_boost        int default 0,
  add column is_favorite      boolean default false,
  add column match_score       int;     -- least(100, base_score + tier_boost)

```

Read model returned to the client

```

GET /api/taxonomy → {
  counts: { groups: 3, roles: 27, titles: 484, favorites: 84 },
  dirty: 0,                                     // draft rows differing from published
  groups: [{
    id, slug: "csuite", name: "C Suite", seniority: "C-level",
    folder: { id, path: "Jobs/C-Suite" },
    rule_text: null,
    roles: [{
      id, slug: "cto", name: "CTO",
      folder: { id, path: "Jobs/CTO" },
      rule_text: null,
      baseline_id: "...",                // null → "Create baseline" affordance
      stats: { titles: 22, fav: 6, watch: 16, off: 0 },
      titles: [{ id, label: "Platform CTO", tier: "fav", live: 3 }]
      //                                     ↑ tier = draft ?? published
      //                                     live = open opportunities matched to this title
    }]
  }]
}

```

`live` is the per-title count of currently-open opportunities matched to that variant — rendered as "3 live" on the row. In the prototype it is a deterministic stub (`hashN(role.id + title, 5)`); wire it to `count(opportunity where matched_title_id = ...)`.

4. Seed data

The full $3 \times 27 \times 484$ seed is machine-readable in `proto-compass/taxonomy.js` (`ROLE_TAX`) — import it directly rather than retyping. Shape per role: `{ id, name, folder, titles: string[], fav: string[], qualifier?: string }`. `fav` holds the seeded favorites (84 across the taxonomy, ~3–6 per role); everything else seeds as `watch`; nothing seeds as `off`.

Seeded folder convention: `Jobs/<ROLE>` per role (`Jobs/CTO`, `Jobs/VP-Engineering`, `Jobs/Dir-Data-AI`), `Jobs/<GROUP>` as the group fallback. Folders must be resolved against the user's real folder list at setup time — do not assume these exist.

5. Matching & normalization pipeline

```
alert ingested (mail_folder watch)
  → extract candidate title strings from subject + body
  → normalize each candidate
  → resolve against title_variant.normalized
  → apply inclusion rule for the matched role/group
  → tier gate
  → score
  → route

normalize(s):
  lowercase
  cut trailing context at the first of | . - , " at " " @ "
  '&' → ' and ' ; strip punctuation ; collapse whitespace
  expand abbreviations (bidirectional dictionary, longest-match first):
    cto→chief technology officer      cio→chief information officer
    cdo→chief data officer | chief digital officer  (ambiguous: keep both candidates)
    cpo→chief product officer          caio→chief ai officer
    coo→chief operating officer        vp→vice president
    svp/evp→senior vice president / executive vice president
    ai→artificial intelligence (keep both forms)    ml→machine learning
    ea→enterprise architecture          it→information technology
  drop filler: "of", "the", "for", "global" is KEPT (it is a real qualifier)

resolve(normalized):
  1. exact match on title_variant.normalized          → method 'exact',      confidence 1.00
  2. exact match on an expanded alias of the candidate → method 'normalized', confidence 0.95
  3. trigram similarity ≥ 0.82 (pg_trgm), highest wins → method 'fuzzy',      confidence =
similarity
  4. else → role-level keyword fallback, tier = 'watch' → method 'none',      confidence 0.00
```

Inclusion rules

Stored as `jsonb` on role/group and evaluated after resolve. Two rules ship seeded:

```
// role 'coo' (C Suite → C00)
{ "mode": "require_any_keyword",
  "keywords": ["software","digital transformation","product delivery",
              "technology operations","business transformation"],
  "scope": "jd_body+title" }

// group 'dir' (Director)
{ "mode": "require_seniority_or_exception",
  "seniority": ["senior director","executive director","managing director","global director"],
  "exception_any": { "comp_min": 400000, "team_min": 100,
                    "scope": "enterprise", "succession": true } }
```

Failing a rule does **not** discard the alert — it records `title_match.rule_passed = false` with a reason and routes the opportunity to the role's *backlog* instead of the morning queue. The user can still find it under the role, filtered.

Tier gate & scoring

TIER	INGEST	SCORE	QUEUE	NOTIFY	PACKET
★ favorite	yes	base + 15 , capped 100	pinned above the rest of the role bin	push immediately	auto-start draft from role baseline
watch	yes	base	normal order (score desc)	daily digest	manual
off	no — dropped at ingest, logged in <code>title_match</code> only	—	—	—	—

`FAVORITE_BOOST = 15` . Keep it a single named constant — the whole point is that one favorite title outranks a higher-paying non-favorite. `is_favorite = true` drives the gold star on cards and pipeline rows.

6. API surface

METHOD	PATH	BODY / NOTES
GET	<code>/api/taxonomy</code>	Full read model (§3). Cache per user; invalidate on any write below.
PATCH	<code>/api/titles/:id/tier</code>	<code>{ tier: 'fav' 'watch' 'off' }</code> → upsert <code>title_tier_draft</code> . Returns new <code>dirty</code> count.
POST	<code>/api/roles/:id/titles/bulk-tier</code>	<code>{ tier, title_ids?: uuid[] }</code> — omit <code>title_ids</code> to hit every title in the role. Backs "★ Favorite all / Watch all / Turn role off". Must be one atomic write (see gap G4).
POST	<code>/api/taxonomy/publish</code>	Copy drafts → <code>title_variant.tier</code> , clear drafts, emit <code>taxonomy.published</code> . Backs "Save favorites".
POST	<code>/api/taxonomy/revert</code>	Delete this user's drafts. Backs "Revert N".
POST	<code>/api/role-groups</code> · <code>/api/roles</code>	<code>{ name, seniority?, folder_id?, rule?, group_id }</code> . Backs "+ Add role group" / "+ Add role".
PUT	<code>/api/roles/:id/folder</code>	<code>{ folder_id }</code> . Backs the folder "Change" control (currently deep-links to intake setup).

GET	/api/mail/folders	Folder tree for the active connection – populates the folder picker.
POST	/api/roles/:id/baseline	Creates an empty baseline. Backs "Create baseline".

Events

```
taxonomy.published { user_id, changed_title_ids[] }
  → job rescore_opportunities(user_id)
    recompute tier_boost / is_favorite / match_score for every OPEN opportunity
    opportunities whose title went 'off' → archive, remove from queue
    opportunities whose title went 'fav' → pin + push notification + packet autostart
  → job rebuild_folder_watches(user_id) // folder bindings may have changed

alert.ingested { alert_id } → match_title → opportunity.created { is_favorite }
opportunity.created(is_favorite=true) → notify.push (bypasses digest batching)
```

7. UI specification

Layout & breakpoints

WIDTH	LAYOUT	GRID
≥ 1180px	Three panes, full height, each pane scrolls internally	220px minmax(360px, 1fr) minmax(270px, 320px), areas "tree list detail"
720–1179px	Tree + title list side by side (internal scroll); detail pane below as auto-fit columns; page scrolls	200px minmax(0, 1fr) / rows minmax(400px, calc(100vh - 230px)) auto
< 720px	Fully stacked, page scrolls. Only the active group is expanded so the tree stays short.	single column

Panes

Header (sticky, above the panes)

Title + one-line explainer on the left; four counters right-aligned (Groups · Roles · Titles · **Favorites** in gold #c08a1e); then Revert N (ghost, only when dirty > 0) and Save favorites (accent). Both buttons white-space:nowrap; flex-shrink:0.

Pane 1 — tree data-group / data-role

Group header row: caret ▼/► · name (700, .02em) · gold ★N total. Role row: indented 28px, name · gold ★N (hidden at 0) · total title count. Selected role = teal fill, white text. Footer: + Add role, + Add role group (ghost, full width).

Pane 2 — title variants

Header: {Group} → {Role} + stats line "22 title variants · 6 favorite · 22 watching · 0 off"; search input; filter chips All · ★ Favorites · Watching · Off (active = ink fill) and a Favorites first toggle (active = teal fill); then the inclusion-rule note (role rule wins over group rule) as a Compass info note.

Row (28px, 1px rule): star button · title label (600 when favorite, strikethrough when off) · N live · tier label (click cycles). Favorite rows get a rgba(192,138,30,.06) wash; off rows drop to 50% opacity. Footer bulk bar: ★ Favorite all · Watch all · Turn role off + the hint "Click the tier label to cycle favorite → watching → off".

Pane 3 — role detail

Five stacked cards: Role (name + group · seniority) · Watched inbox folder (mono path + Change + fallback line) · Favorites in this role (count / total, gold progress bar, up to 6 chips + "+N more") · How favorites get promoted (the five promotion rules from §5) · Role baseline (Open baseline → when present, else Create baseline).

DOM hooks (build against these)

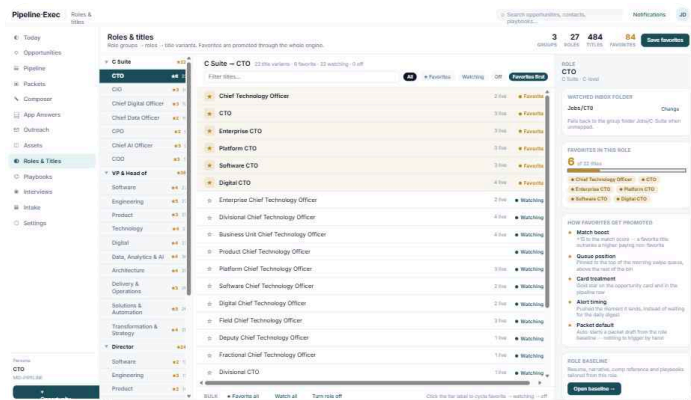
SELECTOR	ELEMENT	REFLECTED STATE
[data-group="csuite"]	Group header (click toggles)	data-open="true false"
[data-role="cto"]	Role row	data-selected="true false"
[data-title="Platform CTO"]	Title row	data-tier="fav watch off"
[data-title="..."] [data-star]	Star toggle (favorite ⇌ watching)	data-star="on off"
[data-title="..."] [data-tiercycle]	Tier label (cycles fav → watch → off)	—
[data-search]	Title filter input	value
[data-filter="all fav watch off"]	Filter chip	data-on="true false"
[data-favfirst]	Favorites-first sort toggle	data-favfirst="true false"
[data-bulk="fav watch off"]	Bulk action	—
[data-action="save revert"]	Publish / discard drafts	revert present only when dirty
[data-action="folder-change"]	Folder picker entry → #/intake/setup	—
[data-action="baseline create-baseline"]	Baseline link / create	—
[data-action="add-role add-group"]	Taxonomy creation	—

Local state (client)

```
groupId, roleId          // selection
q                        // title search, cleared on role change
filter: all|fav|watch|off
favFirst: boolean        // default true – sort rank fav 0, watch 1, off 2
collapsed: {groupId:bool} // wide: all groups open; narrow: only active group open
draft: {"<roleSlug>::<title>": tier} // working set → title_tier_draft
saved: {"<roleSlug>::<title>": tier} // published → title_variant.tier
dirty = |{k : draft[k] ≡ saved[k]}|
tierOf(role,title) = draft[k] ?? (seedFavorites.includes(title) ? 'fav' : 'watch')
```

8. Screen states — every click-through

20 states, in click order. **Trigger** is the literal DOM action; **Result** is what must change.



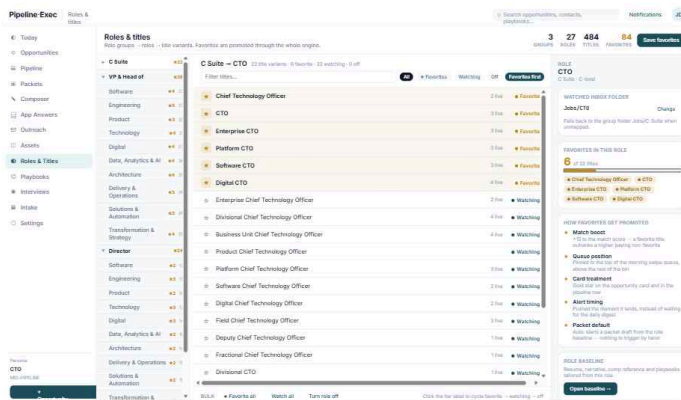
R-1 · DEFAULT

Landing state, three panes

ROUTE `#/roles`

TRIGGER Left nav → "Roles & Titles"

RESULT C Suite → CTO preselected; all groups expanded; Favorites-first on; header counters 3 / 27 / 484 / 84; no Revert chip.

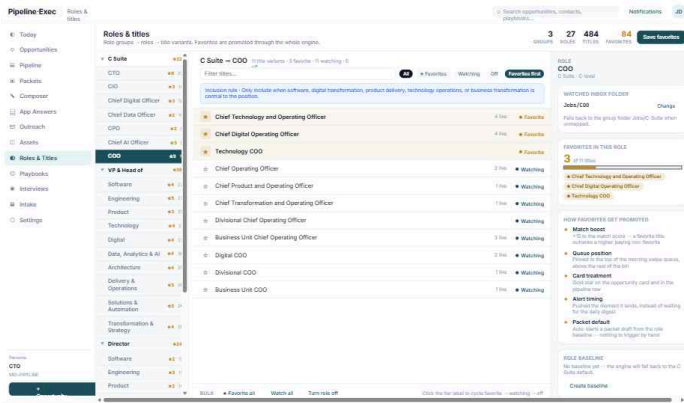


R-2 · TREE

Group collapsed

TRIGGER `[data-group="csuite"]`

RESULT Caret → `▶`, roles hidden, `data-open="false"`. Selection and the other two panes are unchanged (collapsing is not deselecting).



R-3 · RULE

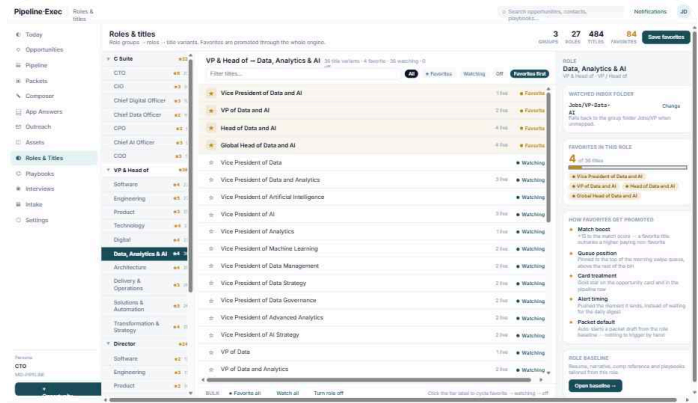
Role with an inclusion rule

TRIGGER

[data-role="coo"]

RESULT

Info note "Inclusion rule · Only include when software, digital transformation, product delivery, technology operations, or business transformation is central to the position." 11 titles, 3 favorite. Detail pane switches to COO / Jobs/COO , no baseline → Create baseline.



R-4 · GROUP SWITCH

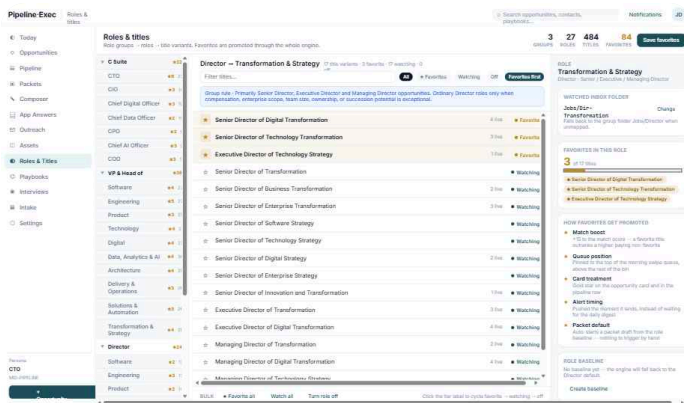
Different group, different role

TRIGGER

[data-group="vp"] then [data-role="vp-data"]

RESULT

Header "VP & Head of → Data, Analytics & AI", 36 variants / 4 favorite; folder Jobs/VP-Data-AI ; search cleared.



R-5 · GROUP RULE

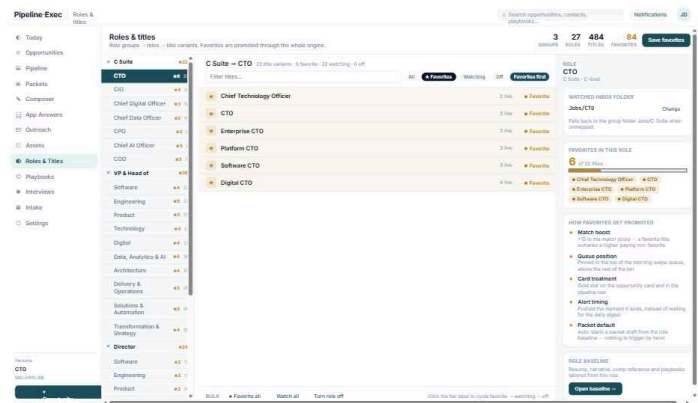
Group-level rule inherited

TRIGGER

[data-group="dir"] then [data-role="dir-transform"]

RESULT

Note prefixed "Group rule -" (role has no own rule): Senior/Executive/Managing Director only, ordinary Director on exception. Seniority line reads "Director · Senior / Executive / Managing Director".



R-6 · FILTER

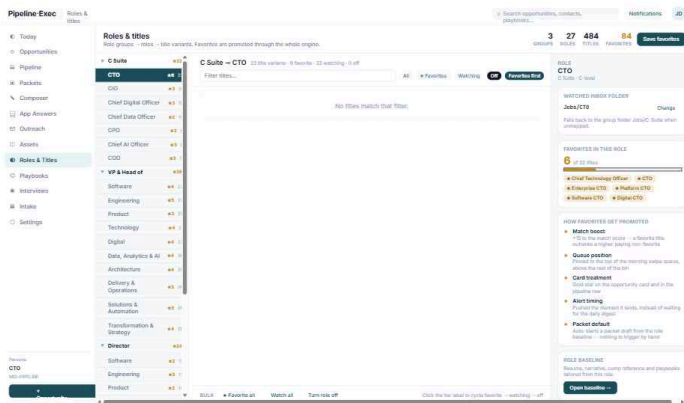
Filter: ★ Favorites

TRIGGER

[data-filter="fav"]

RESULT

List reduced to the role's favorites only; chip fills ink; stats line unchanged (it describes the role, not the filter).



R-7 · EMPTY

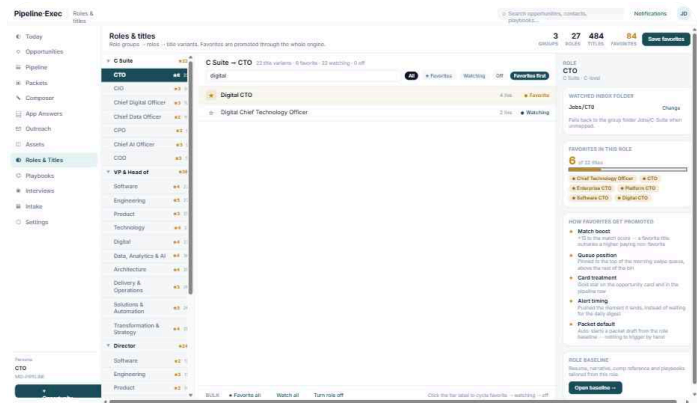
Filter with no matches

TRIGGER

`[data-filter="off"]` with off-titles

RESULT

Dashed empty state "No titles match that filter." Bulk bar stays enabled.



R-8 · SEARCH

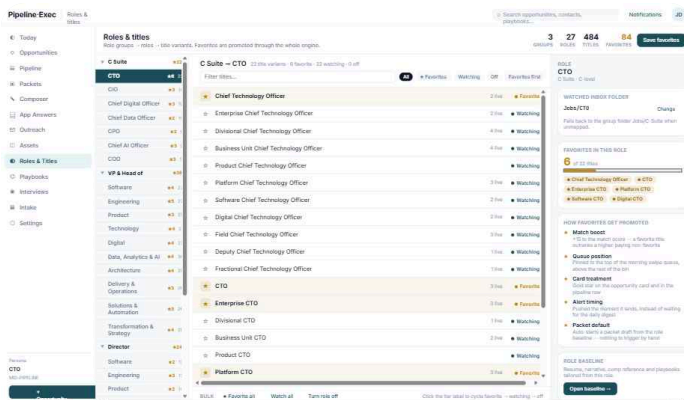
Title search

TRIGGER

`[data-search] = "digital"`

RESULT

Case-insensitive substring filter within the selected role only, combined with the tier filter. Cleared automatically when the role changes.



R-9 · SORT

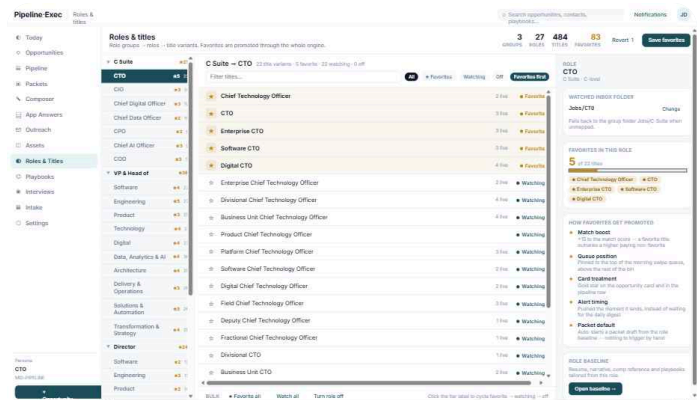
Favorites-first off

TRIGGER

`[data-favfirst]`

RESULT

Rows return to seed order (long forms then abbreviations); toggle loses its teal fill. Rank when on: fav 0, watch 1, off 2, stable within tier.



R-10 · DEMOTE

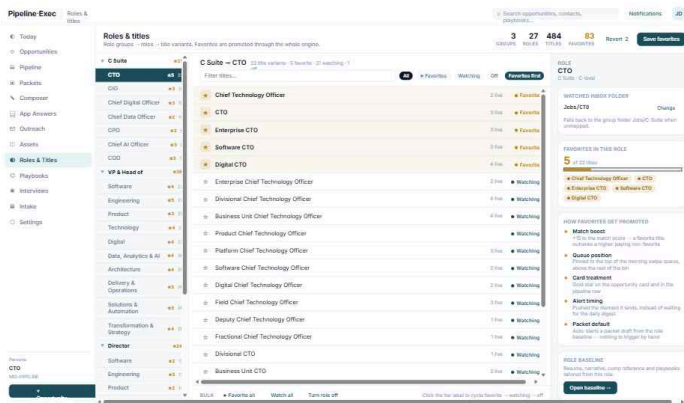
Star toggled off → draft dirty

TRIGGER

`[data-title="Platform CTO"]`

RESULT

Row → `data-tier="watch"`, gold wash and bold drop, favorites 6→5 in both the stats line and the detail card, header Favorites 84→83, **Revert 1** appears.



R-11 · CYCLE

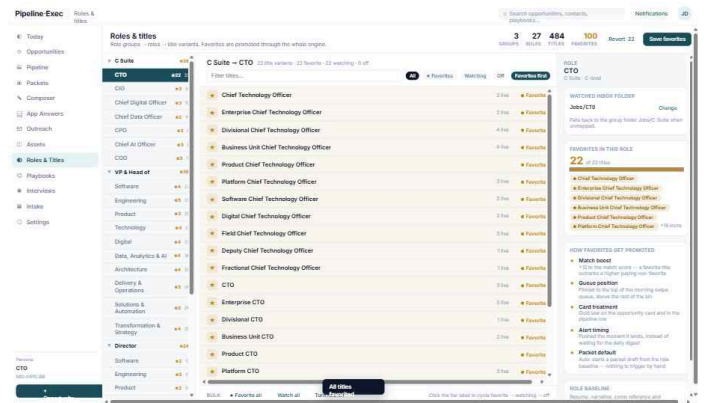
Tier cycled to Off

TRIGGER

[data-title="Field CTO"] [data-tiercycle]

RESULT

watching → off: strikethrough label, 50% opacity, grey dot; stats "...21 watching · 1 off". Cycle order is fav → watch → off → fav.



R-12 · BULK

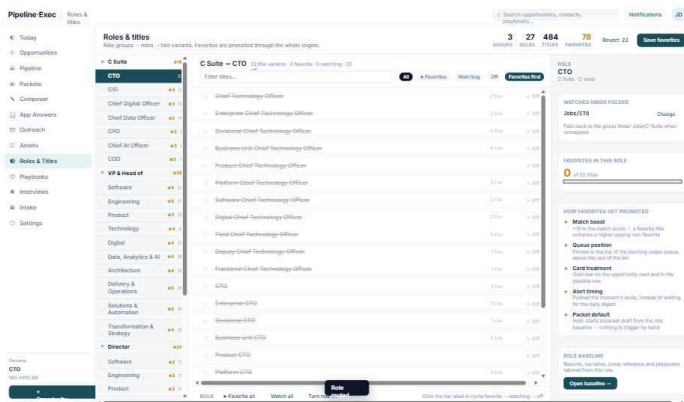
★ Favorite all

TRIGGER

[data-bulk="fav"]

RESULT

All 22 titles → favorite in one write, toast "All titles favorited", header Favorites 100, Revert 22, detail card 22/22 with "+16 more".



R-13 · BULK

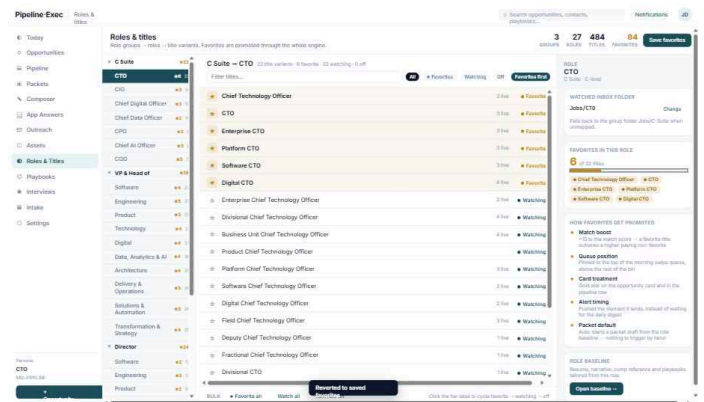
Turn role off

TRIGGER

[data-bulk="off"]

RESULT

Every title struck through, toast "Role muted", favorites 0. Server-side this stops ingestion for the whole role.



R-14 · REVERT

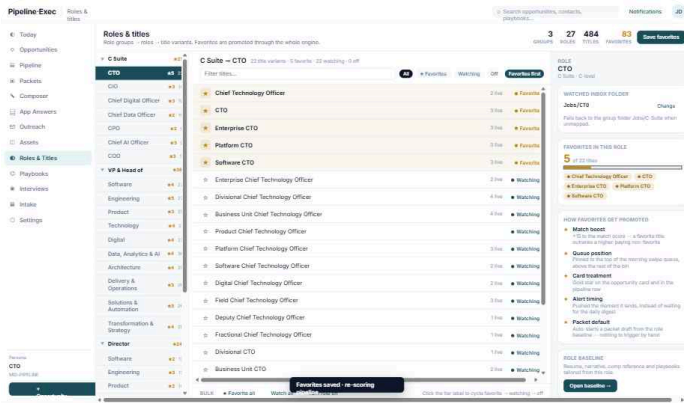
Revert drafts

TRIGGER

[data-action="revert"]

RESULT

Drafts discarded → back to the published set (seeded 6 favorites), toast "Reverted to saved favorites", Revert chip disappears.



R-15 · PUBLISH

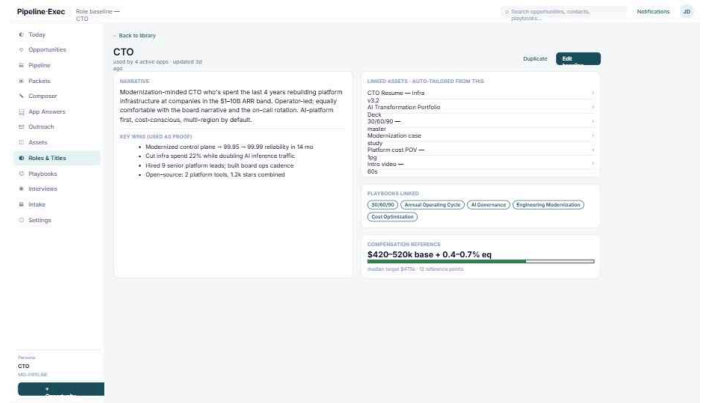
Save favorites

TRIGGER

[data-action="save"]

RESULT

Drafts published, dirty → 0, Revert chip gone, toast "Favorites saved · re-scoring pipeline". Emits `taxonomy.published` → rescore job.



R-16 · BASELINE

Role baseline detail

ROUTE

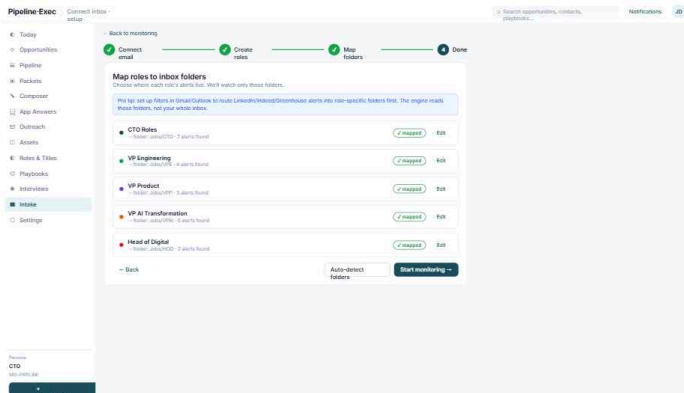
`#/roles/baseline/cto`

TRIGGER

[data-action="baseline"]

RESULT

Existing baseline screen: narrative, key wins, linked assets, playbooks, comp reference. "← Back to library" returns to `#/roles`.



R-17 · FOLDER

Change watched folder

ROUTE

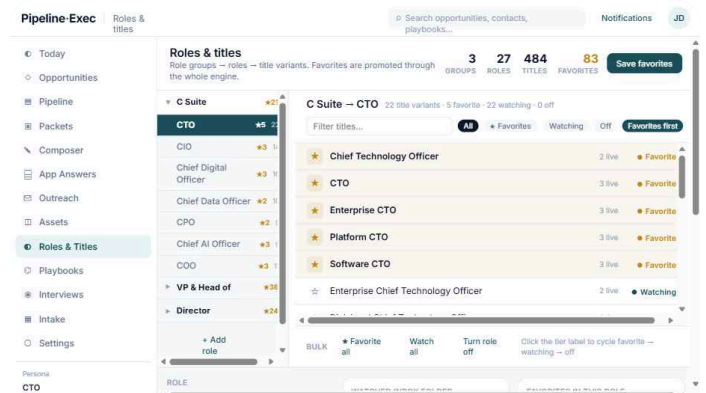
`#/intake/setup`

TRIGGER

[data-action="folder-change"]

RESULT

Jumps to the inbox setup stepper on step 3 "Map folders". **Gap G1**: this list is still the legacy five families and is not generated from the taxonomy — it must render one row per role, with a real folder picker.



R-18 · RESPONSIVE

Two-pane (720–1179px)

TRIGGER

Viewport 1000×580

RESULT

Tree + list side by side with internal scroll; detail pane below as auto-fit 240px columns; page scrolls.

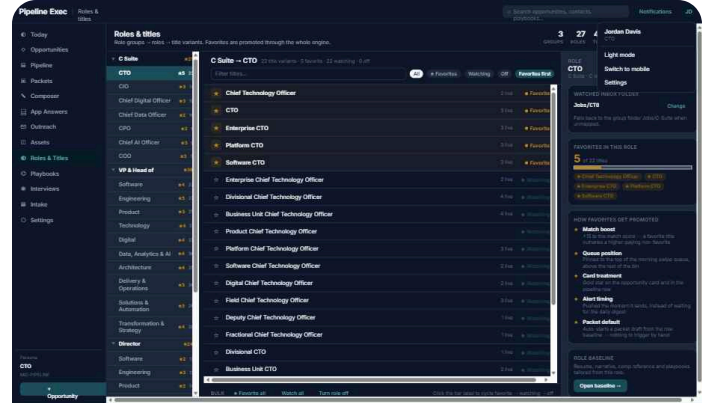


R-19 · RESPONSIVE

Stacked (< 720px)

TRIGGER Viewport 620×940

RESULT Single column: tree (only the active group expanded), then list, then detail. Counters wrap under the title.



R-20 · THEME

Dark mode (avatar menu)

TRIGGER Avatar → "Dark mode"

RESULT Compass dark tokens; gold favorite accent and teal selection both hold contrast. Menu items: Light mode · Switch to mobile · Settings.

9. Acceptance criteria

- Seed import produces exactly 3 groups, 27 roles, 484 titles, 84 favorites; counters match.
- Every control in the §7 hook table exists with that selector and reflects its state attribute.
- Star toggle switches favorite ⇌ watching only; the tier label cycles all three.
- Bulk actions are a single atomic write — after "★ Favorite all" on CTO, all 22 rows read `data-tier="fav"`.
- Save clears dirty and hides Revert; Revert restores the published set and hides itself.
- An alert whose title resolves to an `off` variant never creates an opportunity.
- Two otherwise-identical opportunities differing only in favorite status rank favorite-first, with a 15-point score gap.
- A COO alert with no transformation/software signal lands in backlog with `rule_passed = false`, not in the morning queue.
- All three breakpoints render without clipping; no horizontal scroll inside the page body.

10. Known gaps in the reference implementation

G1 · Folder mapping is not taxonomy-driven. `#/intake/setup` still lists five legacy role families from `intake.jsx` (`FAMILY_FOLDER`) with static "✓ mapped" pills. Rebuild it from `role` rows with a real folder picker fed by `GET /api/mail/folders` .

G2 · Promotion is specified, not wired. The five promotion rules render in the detail pane but nothing yet applies `+15` , pins the queue, stars the card, or autostarts a packet. Scoring, swipe order, and the opportunity/pipeline rows are untouched.

G3 · "+ Add role" / "+ Add role group" are toasts. No creation form; the taxonomy is seed-only. Needs name, group, folder, optional inclusion rule, and title-variant entry (paste-a-list is the fastest input for this data shape).

G4 · Seeded favorites are a guess. The 84 seeded favorites were inferred from a platform/AI/transformation profile, not chosen by the user. Treat them as a starting point and surface an onboarding pass that walks role by role.

G5 · live counts are stubs. Deterministic hash, not real opportunity counts.

G6 · Tiers persist to localStorage. Two keys (`ee-role-tiers-v1` working, `ee-role-tiers-saved-v1` published) stand in for the draft/publish tables.